

Executive Framework Document

When Maintenance Data Disagrees

A practitioner framework for AI in rail asset management

Prepared by: Sid Pandey · August 2026

*First presented as a guest lecture to students and rail industry professionals
Independent research. No employer data or systems.*

Executives and asset custodians often struggle to act confidently because an asset's condition changes depending on which system they consult. The problem is not always missing data, but fragmented and conflicting evidence across Maximo, inspections, SCADA/IoT, ERP etc. This framework uses AI to connect those records, propose reconciliations or abstain, while a named human decides and Maximo remains the system of record. The payoff is more complete, traceable and trustworthy information for analysis and decisions. Organisations do not need perfect data; they need governed, decision-ready evidence.

Different Systems. One Failure. Different Different Stories.

When maintenance data is described as incomplete, the information is often not entirely absent; it is distributed across systems. The four sources shown here are illustrative, the framework can bring together as many relevant, governed data sources as the use case requires. Each source contributes a partial view of the same asset event.



Maximo CMMS

Recorded: "General defect"

Broad category, no specificity



Inspection Narrative

Recorded: "Bearing noise"

Technician observation, unstructured text



SCADA / IoT

Recorded: "Vibration spike (pre-event)"

Sensor data, timestamped



Materials / ERP

Recorded: "Seal kit issued"

Parts transaction, no fault linkage

Same asset. Multiple systems. Partial records. Linked together, they create the most complete evidence-backed view available.

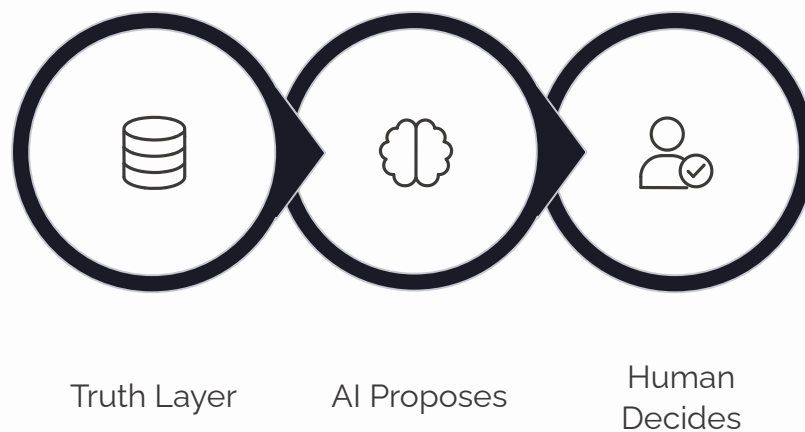


What this means: The goal is to reconcile these four partial truths: Maximo, SCADA/IoT, inspection narratives, and Materials/ERP, into a single trusted picture of asset condition before any AI prediction or maintenance decision is made.

Illustrative configuration and record. The number and type of source systems will vary by operator, asset class and decision.

The Core Principle: Establish Trust Before Prediction

The most common AI failure in maintenance is applying prediction to data that hasn't been reconciled. This framework enforces a strict sequence: first establish a trusted, evidence-linked view of asset condition; then let AI propose ranked resolutions; then a named human reviewer makes every consequential call.



⊗ Skipping this sequence means confident predictions on unresolved conflicts.

Truth Layer

Evidence linked and reconciled across all four systems: Maximo, SCADA/IoT, inspection narratives, and Materials/ERP, before any prediction is attempted.

AI Proposes

Ranked resolution options generated by the model, each accompanied by the supporting evidence behind it so reviewers can judge the basis of any recommendation.

Human Decides

A named person is accountable for every consequential call. No action is taken on AI output alone, and the decision trail is explicit and auditable.

AI Boundaries: What It Can and Cannot Do

Hard limits on what AI can do are not a weakness, they are what makes the system trustworthy. The boundaries below are non-negotiable design constraints, not temporary restrictions. They exist because unconstrained AI in safety-critical environments creates liability, not value.

AUTHORISED



Extract Structure

Pull organised data from unstructured maintenance records



Map Taxonomy

Align asset classifications across systems



Reconcile Conflicts

Flag and rank discrepancies for human review



Retrieve Evidence

Surface supporting records linked to a query



Abstain / Return Unknown

Declare uncertainty rather than guess

NEVER AUTHORISED



Approve Own Correction

The model cannot validate its own output



Hold Maximo Write Credential

No direct database access, ever



Change Safety Fields Automatically

Safety records require human sign-off



Publish KPI Without Lineage

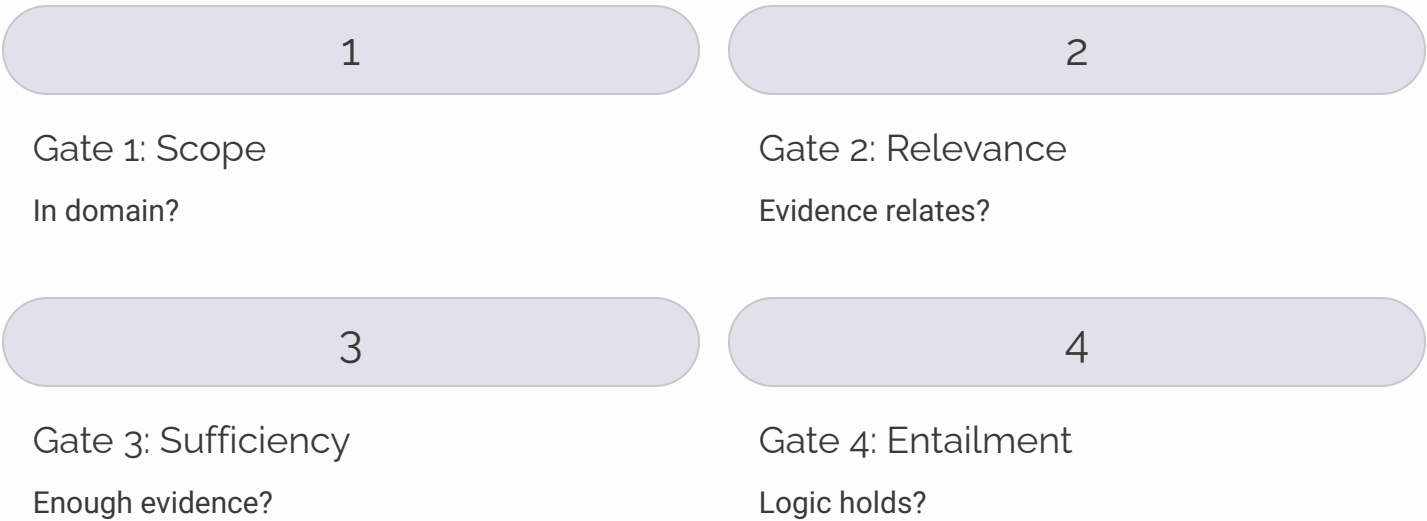
Every metric must show its evidence chain



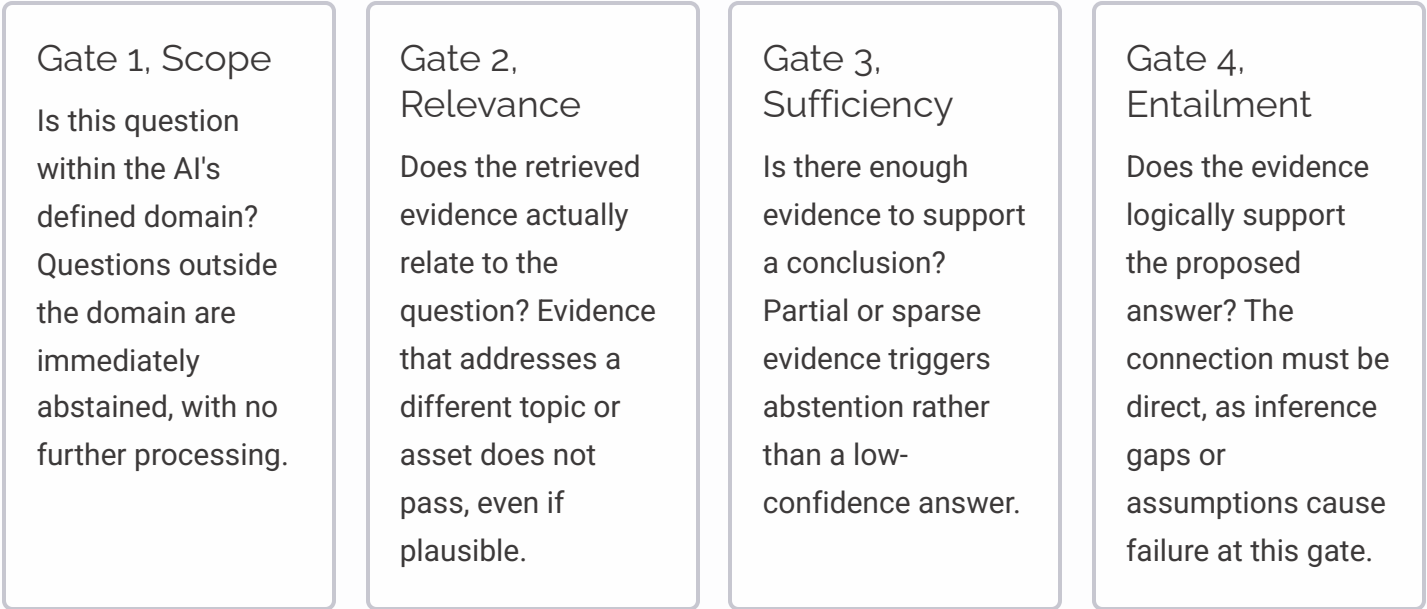
Unknown is a valid result. Unsupported confidence is a design failure.

The Four-Gate Safety Check

A single AI confidence score proved unreliable in prototype testing: it could be high even when the underlying evidence was weak. The framework replaced it with four independent gates. Every AI response must pass all four sequentially. A single failure at any gate routes the output to 'Abstain / Unknown' rather than producing a potentially misleading answer.

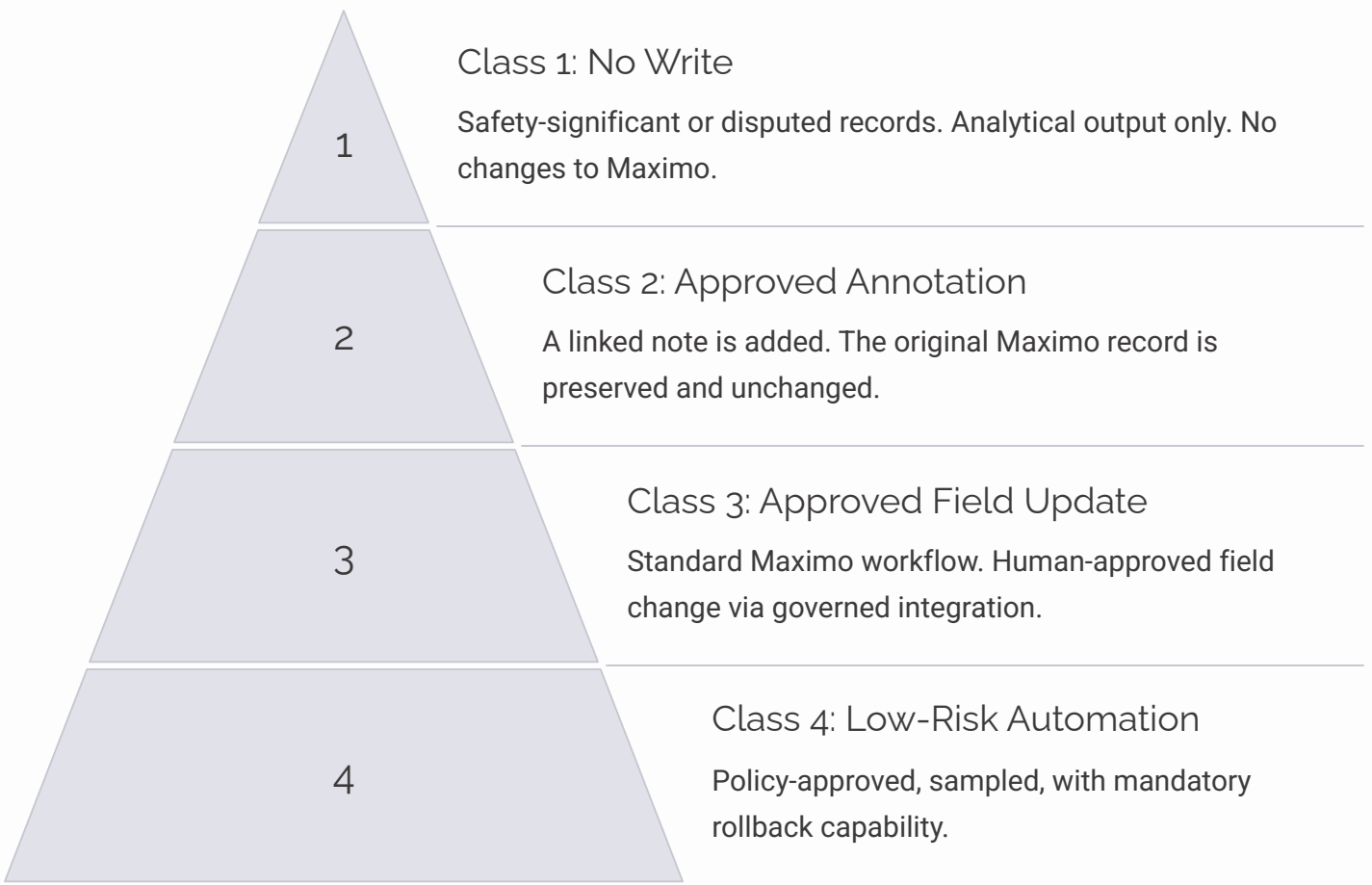


Any gate failure routes to Abstain / Unknown. All four must pass for output to proceed.



Write-Back Governance: Four Strictly Controlled Classes

The AI model never holds a credential that can directly update Maximo, the asset management system of record. All data changes flow through a governed integration service with human approval at each level. The four classes below define what type of change is permitted, and under what conditions.



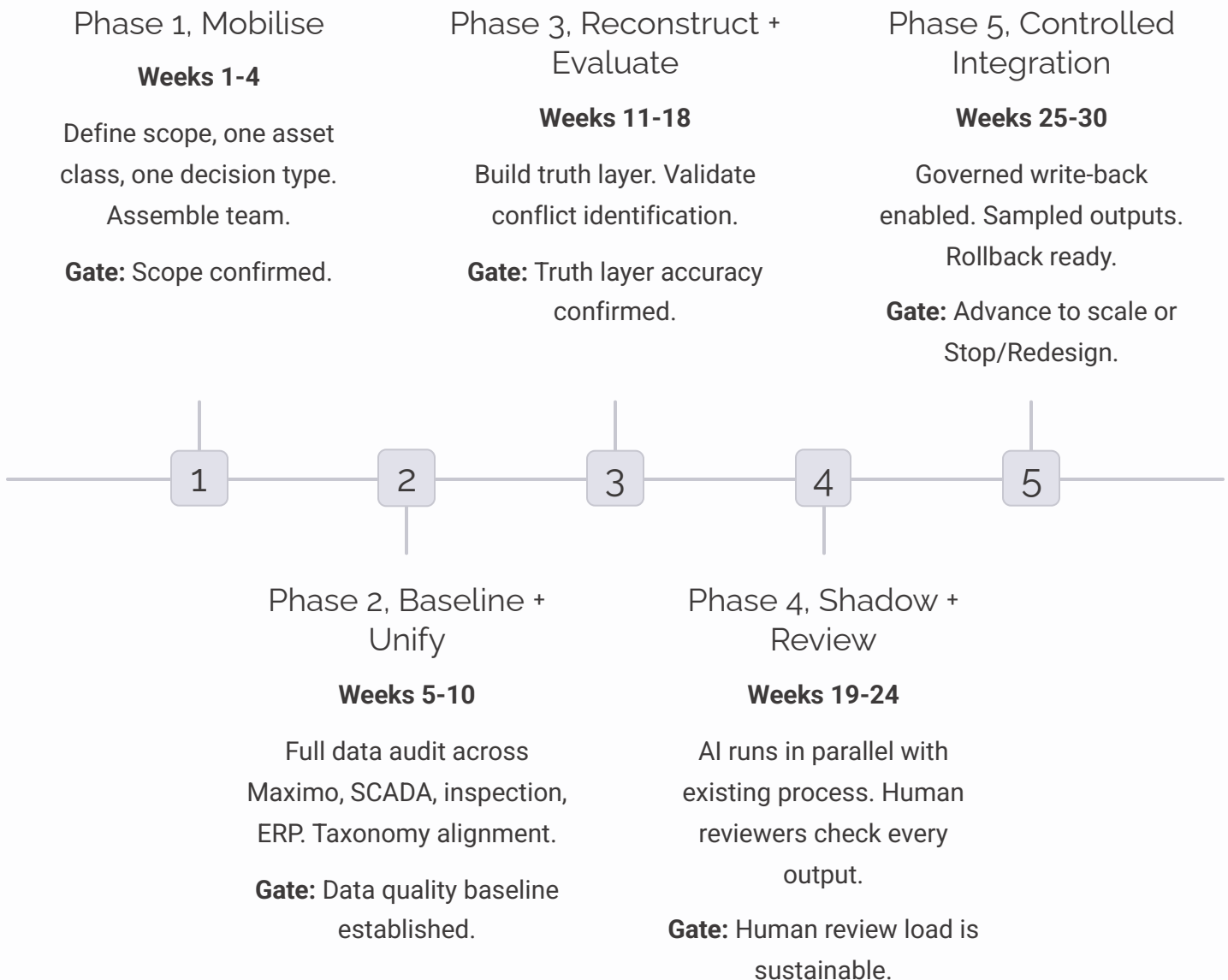
Class 1 is the default. Any disputed or safety-significant record stays read-only unless explicitly escalated by a human reviewer.

 **No direct Maximo write credential. Ever.**

26-30 Week Pilot: Earn the Right to Scale

26 TO 30 WEEKS · 5 GATED PHASES · ONE ASSET CLASS

The pilot is deliberately narrow: one asset class, one decision type. Scope is constrained before schedule is committed. Each phase ends with a gate: **Advance** (evidence supports continuation) or **Stop / Redesign** (return to previous phase). No phase is skipped. The right to scale is earned by evidence, not assumed.



Gate criteria are defined before the pilot begins, not after results are seen. This prevents confirmation bias from driving the decision to continue.

The Operating Architecture at a Glance

Maximo is not replaced - it remains the system of record throughout. The architecture adds a governed layer above it: raw operational data flows up through six stages, AI services operate only within a tightly bounded middle layer, and all approved changes return to Maximo through a controlled integration service. The diagram below shows how data moves through the system.

		
Operational Evidence Raw data from Maximo CMMS, SCADA/IoT, Inspection Narratives, and Materials/ERP. Multiple partial views of the same asset.	Governed Integration Layer Data policy enforcement, RAAC access controls, and full lineage tracking applied before any processing.	Conflict Identification AI flags discrepancies across sources. No resolution is attempted at this stage.
		
Narrow AI Services Evidence is ranked and resolution proposals are generated. The model abstains when evidence is insufficient.	Human Adjudication A named reviewer evaluates every proposal. No consequential decision is taken without human sign-off.	Trusted Data & Decisions Approved changes return to Maximo only via the governed integration service. The system of record is never bypassed.

 No direct production credential. Approved writes return through governed integration service only. Maximo remains the system of record, always.

Prototype Signals: What the Evidence Shows

The following results come from prototype testing on open and synthetic data, not production rail systems. They are directional signals that indicate the framework is worth pursuing, not production claims. Each figure requires a reproducibility pack before it can be used to justify investment decisions.



Why this matters: By recovering and reconciling evidence across systems, teams can undertake reliability analysis, maintenance planning and investment prioritisation without waiting for every source record to be complete.

903→91

Asset Register

Candidates isolated for
engineering review

99%

Auto-Reconciled

Operational timetable records
resolved

5–14days

Early Warning

Failure warnings ahead of
documented breakdowns

These signals informed the decision to proceed to a controlled pilot, they did not replace it.



Prototype signals only. Reproducibility pack required before investment reliance.

The Questions That Matter Most

This framework is designed to be challenged, not accepted. The four questions below are the ones most likely to reveal where it holds and where it needs adaptation for your specific operating context. If you are reviewing this document ahead of a conversation, these are the areas where your experience will add the most value.



Source Conflict Burden

Which system conflicts create the greatest review burden?



Human Review in Practice

Where has human review proved non-negotiable?



Integration Reality

Which integrations and taxonomies have worked or failed?



Credibility Threshold

What evidence would you need before trusting this near production?



Responses to these questions will shape the pilot scope definition in Phase 1 (Mobilise).

References and Further Reading

This document draws on the following standards, datasets, and published research. All sources are publicly available unless noted.

Standards and Frameworks

- **ISO 55001:2024:** Asset management. Management systems. Requirements.
- **ISO 14224:** Petroleum, petrochemical and natural gas industries. Collection and exchange of reliability and maintenance data for equipment.
- **NIST AI RMF 1.0:** Artificial Intelligence Risk Management Framework. National Institute of Standards and Technology, 2023.
- **NIST AI 600-1:** Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile. NIST, 2024.
- Guidance for AI Adoption (National AI Centre, 2025).
- IBM Maximo Manage documentation (current release).

Datasets and Research

- Vicmap Transport (open dataset, Victoria, Australia).
- **PTV GTFS:** Public Transport Victoria General Transit Feed Specification data.
- OpenStreetMap contributors.
- **MetroPT-3 dataset:** UCI Machine Learning Repository, dataset 791.
- Brundage, M. P., Sexton, Hodkiewicz, Dima, Lukens. Technical Language Processing: Unlocking Maintenance Knowledge, Manufacturing Letters (2020).
- *Stewart, M., Hodkiewicz, M., and Li, S. Large Language Models for Failure Mode Classification: An Investigation (2023, arXiv:2309.08181).*
- Lutz, M.-A. et al. KPI Extraction from Maintenance Work Orders: A Comparison of Expert Labeling, Text Classification and AI-Assisted Tagging (2023, arXiv:2311.04064).

Full paper: *The Work Order Truth Framework, v1.1, available on request.*

Sid Pandey · siddhartha27@gmail.com · linkedin.com/in/sidpandey6 · Independent research. No employer data or systems.